Case 8 — Ms. Chloe Davies

27 F · Functional Neurological Disorder (FND) — Motor Subtype | Left leg drag / weakness Left arm tremor Inconsistent on testing Normal MRI & NCS Precipitant: Psychological stress Positive Hoover sign

Neurological Physiotherapy

■ SUBJECTIVE PROBLEM LIST

- Physical Impairments
- Left leg weakness — inconsistent, attention-modulated
- Abnormal gait — left foot drag pattern
- Possible secondary deconditioning from activity avoidance
- Difficulty walking safely and efficiently in the community
- Reduced independence in ADLs and community participation

■■ RED FLAGS

- Clinical Significance
- Action if Present
- Suicidal ideation or severe depression
- FND has a high burden of psychological comorbidity and significantly elevated suicide risk in severe or long-standing cases
- Involve mental health team immediately; follow safe messaging guidelines; do not proceed without psychiatric support
- New or evolving neurological symptoms not previously assessed
- Organic pathology can co-exist with FND — a new symptom should never be automatically attributed to FND without re-evaluation by neurology
- Refer back to neurology urgently; document carefully; do not assume all new symptoms are functional

■ PATHOPHYSIOLOGY (KEY POINTS)

- Functional Neurological Disorder is a disorder of neurological function — the brain is producing abnormal motor and sensory output despite structurally intact neural hardware, as confirmed by Chloe's normal MRI and nerve conduction studies.

■ DIFFERENTIAL DIAGNOSES

See full case notes

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Test 1 — Hoover Sign

MEASURES Positive FND motor sign — tests whether lower limb weakness is voluntary (intentional motor drive absent) or associated with intact automatic motor output HOW TO PERFORM Patient lies supine. Explain you are going to test the strength in both legs. Place one hand under the heel of the weak left leg. Ask Chloe to lift the left leg against your resistance — note the force produced (typically reduced in FND).

Test 2 — Tremor Entrainment Test

MEASURES Distinguishes functional tremor (frequency entrains to external rhythm) from organic tremors (fixed frequency driven by a subcortical oscillator) HOW TO PERFORM Ask Chloe to hold the affected left arm outstretched so the tremor is visible. Observe and note the resting tremor frequency. Ask her to tap a specific rhythm with her right (unaffected) hand — start with a slow, clear 2 Hz beat, counting aloud: "one, two, one, two…"

Test 3 — Gait Observation: Inconsistency Assessment

MEASURES

■ TREATMENT PLAN

- "Do the symptoms fluctuate throughout the day? Are there specific situations, emotions, or activities where they are better or worse?"
- "Have you been through anything particularly stressful, traumatic, or emotionally significant around the time this started?"
- "How has this affected your daily life — your work, relationships, getting around, and how you feel about yourself?"
- "Have you seen any other doctors or specialists about this? What did they tell you, and how did that feel?"
- "Do you have a history of any other health conditions, mental health difficulties, or significant past experiences such as trauma?"
- Disclosure of active abuse or safeguarding concern
- Ongoing trauma can be the maintaining factor for FND and poses immediate risk to Chloe's safety
- Follow mandatory safeguarding reporting protocol; involve social work; prioritise safety above therapeutic progress
- Observe Chloe walking from one end of the room to the other with no instruction other than to walk normally — document foot clearance, step length, velocity, and left foot drag.
- Ask her to walk while performing a cognitive dual task: "As you walk, count backwards from 100 in 7s."
- Compare gait quality between conditions — note any improvement with the dual task.
- Observe across multiple trials for intra-session variability — does the gait pattern change when she believes she is not being watched?